MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10548119

(DECODER, RCFD)

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of MIL-C-50652 comprise the complete requirements for the range control fine delay decoder (RCFD decoder) used in Range Computer CP-888/VPS-2, part of Radar Set AN/VPS-2.

2. APPLICABLE DOCUMENTS.

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-50652	Circuit Card Assemblies, General Specification for

DRAWINGS

U.S. Army, Frankford Arsenal

F10548119	Decoder, RCFD

PACKING DATA SHEET

10548119	Packaging of Decoder, RCFD

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(Copies of specifications, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - The assembly shall be manufactured in accordance with Drawing F10548119 and drawings pertaining thereto.

3.2 General specifications. - The assembly shall meet the require - ments of MIL-C-50652.

3.3 Environmental service conditions. - As specified in MIL-C-50652.

3.4 Electrical. - Each conductor shall be electrically continuous as specified on the schematic diagram of the circuit card assembly.

3.5 Performance. - For the requirements of 3.5.3 and 3.5.5, logic input and output levels shall be as follows:

Logic 0: 0.0 ±0.5 volt direct current (Vdc) level.
Logic 1: +3.5 ±1.0 Vdc level.

3.5.1 Input signals. - For all requirements except 3.5.5.2 the input signals to the RCFD decoder shall be T1, T3, $\overline{T1}$ and $\overline{T3}$ as shown in Figure 1. For requirement of 3.5.5.2 a T1 signal as shown in Figure 1 will be alternately applied to P1-41 and 11. A positive pulse shall be a change in direct current (dc) level from 0.0 ±0.5 volt to +3.5 ±1.0 V and return to 0 V. A negative pulse shall be a change in dc level from +3.5 ±1.0 V to 0.0 ±0.5 V and return to +3.5 V.

3.5.2 Input power. - The input power applied between terminals P1-30, 31 and 32 (+5 V) and terminals P1-24, 25 and 26 (+5 V return) shall be 5.0 ±0.3 Vdc, at 300 milliamperes (mA), regulated: with an alternating current (ac) ripple less than 50 millivolts (mV) peak-to-peak.

3.5.3 Velocity output signals. - With input logic levels applied sequentially as specified in Table I, the $\dot{\overline{R}}$ Max, ECM $\dot{R}$ <±5 meters per second (m/s) and $\dot{\overline{R}}$ Min output signals at P1-12, P1-14, and P1-38 respectively shall be as specified in Table I. The out pulses shall have the same characteristics as shown in Figure 1.

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TABLE I. - WAVEFORM OBSERVATIONS

	INPUT TERMINALS				OUTPUT SIGNAL CHARACTERISTICS		
STEP	16	22	50	55	14	12	38
1	1	0	1	1	T3 pulse	$\overline{T1}$ pulse	Logic 1
2	0	1	0	1	T1 pulse	Logic 1	$\overline{T3}$ pulse
3	1	0	1	0	Logic	———	———
4	0	1	0	0	Logic	———	———

3.5.4 RCFD decoder output. - With a logic 0 applied to P1-56, P1-54, P1-44 and P1-52, and signal $\overline{T1}$, as shown in Figure 1, applied to P1-42, 48, 49 and 51; the T1 signal as shown in Figure 1 shall be present at the following terminals: P1-3, P1-7, P1-13, P1-17, P1-5, P1-9, P1-15, P1-21, P1-23, P1-37, P1-43, P1-47, P1-35, P1-39, P1-45, and P1-48. These outputs represent ranges of 18.75 meters (m) through 0 m in 1.25 m increments.

3.5.5 E/L trigger output.

3.5.5.1 E/L output. - With a logic 1 level applied to P1-55 output signal T1 and T3 as shown in Figure 1 shall be present at P1-19. With a logic 0 level applied to P1-55, a logic 0 shall be present at P1-19.

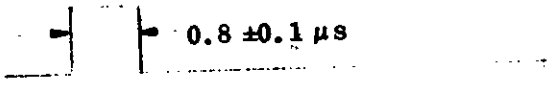
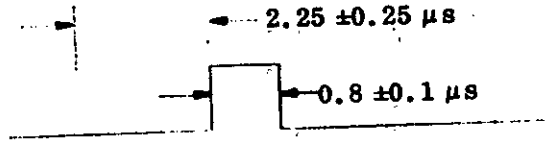
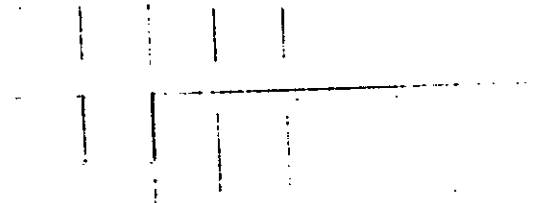
3.5.5.2 E/L delay differential. - With signal T1 as shown in Figure 1 being alternately applied to P1-11 and 41 and a logic 1 level applied to P1-55, the output signal at P1-19 shall be T1 as shown in Figure 1. As the input signal is alternately applied the delay differential of the output signal (the leading edge of each alternate signal) shall show a delay differential of 1.0 nanosecond (ns) maximum.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. - The quality assurance provisions of MIL-C-50652 shall apply to the assembly covered by this detail specification of the circuit card assembly.

4.2 First article inspection. - As specified in MIL-C-50652.

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SIGNAL NAME	WAVESHAPE OR CHARACTERISTIC	TERMINALS
T1		P1-41 (11 - Note 2)
T3		P1-11
T1		P1-18, P1-42, P1-46, P1-49, & P1-51
T3		P1-34
T1 & T3		P1-19

NOTES: 1. All signals at a 20-kHz PRF.
2. T1 applied to P1-11 for 3.5.5.2 only

FIGURE 1. INPUT-OUTPUT SIGNAL CHARACTERISTICS.

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4.3 Inspection provisions. - As specified in MIL-C-50652.

4.4 Special sampling. - As specified in MIL-C-50652.

4.5 Inspection equipment. - As specified in MIL-C-50652 and 4.5.3.

4.5.1 Government-furnished inspection equipment. - As specified in MIL-C-50652.

4.5.2 Contractor-furnished inspection equipment. - As specified in MIL-C-50652.

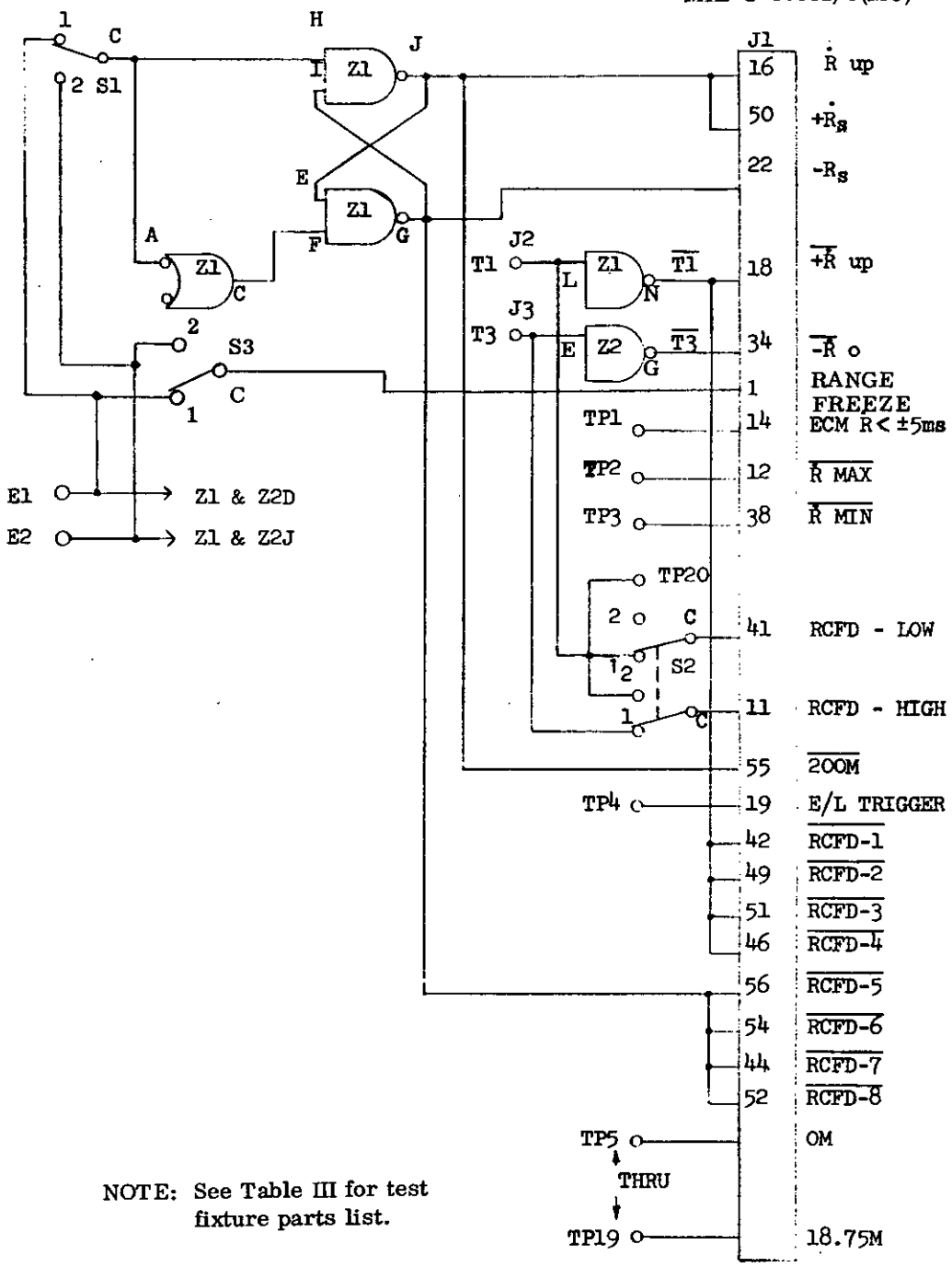
4.5.3 Test equipment. - In conjunction with 4.5 the test equipment listed in Table II shall be utilized in the performance of the applicable tests as specified in 4.6.

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TABLE II. - TEST EQUIPMENT LIST

ITEM NUMBER		DESCRIPTION
1.	Power supply	5.0 ±0.3 Vdc, regulated; ac ripple less than 50 mV peak-to-peak; capacity 300 mA.
2.	Pulse generator	Frequency: 20 kHz minimum Duty factor: 10% minimum Amplitude: minimum of 4 volts into a 50-ohm load. Pulse: Rise fall times less than 0.1 μs.
3.	Pulse generator	Same as item 2.
4.	Oscilloscope	Rise time of scope plus measuring probe 10 ns or less, sweep period 50 ns per division or greater, input resistance greater than 8 megohms, input capacitance less than 25 μF.
5.	Sampling oscilloscope	Rise time of scope plus measuring probe 100 ps or less, sweep period 1 ns per division or greater; input resistance greater than 500 ohms, input capitance less than 10 pF.
6.	Test fixture	As specified in Figure 2. See Table III for test fixture parts list.

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NOTE: See Table III for test fixture parts list.

FIGURE 2. TEST FIXTURE SETUP.

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TABLE III. - TEST FIXTURE PARTS LIST

DESIGNATION	NOMENCLATURE AND DESCRIPTION
E1	Binding post, H. H. Smith 257-R or equivalent.
E2	Binding post, H. H. Smith 257-B or equivalent.
J1	Connector; any of the following or equivalent:
	(1) 5893232-1, Malco Mfg. Co., Inc. Chicago, Illinois.
	(2) 500946-01, National Connector Corp., Minneapolis, Minn.
	(3) 07-5422-0198-01, Elco Corp., Willow Grove, Pa.
J2	Connector, receptacle, UG-25B/U or equivalent.
J3	Same as J2.
S1 and S3	Switch SPDT.
S2	Switch DPDT.
TP1 thru TP20	Tip jack, H. H. Smith 1501 or equivalent.
Z1 and Z2	Integrated circuit, SG-140-03 Sylvania or equivalent.

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4.6 Test methods and procedures.

4.6.1 Environmental. - As specified in MIL-C-50652.

4.6.2 Electrical. - Continuity shall be inspected by standard measuring equipment (SME) and visual examination to determine compliance with 3.4

4.6.3 Performance. - The selected RCFD decoders shall be tested as follows: Set up equipment as shown in Figure 3. Use oscilloscope, item 4, for observation of all input and output signals as specified in 4.6.3.2 through 4.6.3.5.1. Use oscilloscope, item 5, for observation of output signals as specified in 4.6.3.5.2. When specified, a logic 0 shall be observed as a 0.0 ±0.5 Vdc level and a logic 1 as a +3.5 ±1.0 Vdc level. Switch S1 on test fixture, item 6, is used to control logic level inputs to the RCFD assembly. Switch S1, when in position 1, applies logic 0 to P1-16, 50 and 55, and logic 1 to P1-22, 44, 52, 54 and 56. With S1 in position 2 all logic levels are complemented.

4.6.3.1 Input power. - Using item 1, apply input power as specified in 3.5.1.

4.6.3.2 Input signals. - Using pulse generators, items 2 and 3, set up and apply input signals T1 and T3 as shown in Figure 1, to J2 and J3 on test fixture, item 6.

4.6.3.3 Velocity output signals. - At TP1, TP2, and TP3 on the test fixture, item 6, observe the ECM $\dot{R} <$ ±5m/s, $\overline{\dot{R}\ \text{Max}}$ and $\overline{\dot{R}\ \text{Min}}$ output signals as specified in 3.5.3.

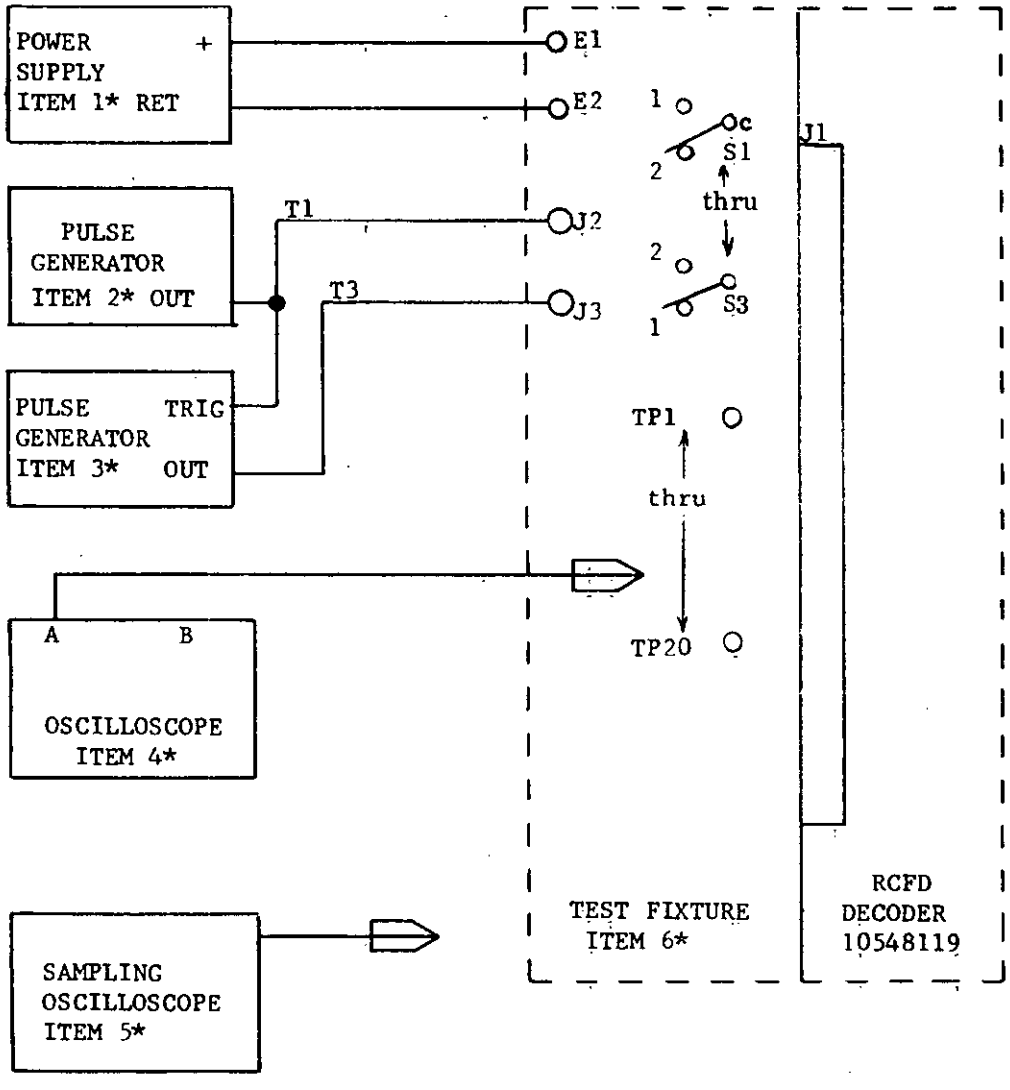
4.6.3.4 RCFD decoder output. - With logic levels applied, as specified in 3.5.4 (S1 in position 2), at TP5 through TP19, observe output signal as specified in 3.5.4.

4.6.3.5 E/L trigger output.

4.6.3.5.1 E/L output. - With a logic 1 level applied as specified in 3.5.5.1, observe at TP4 the output signals as specified in 3.5.5.1.

4.6.3.5.2 E/L delay differential. - Apply signal T1 as specified in 3.5.52 by alternately switching S2 between positions 1 and 2. Using oscilloscope, item 5, observe at TP4 the output signal as specified in 3.5.5.2.

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*REFER TO 4.5.3

FIGURE 3. RCFD DECODER TEST SETUP.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. - Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10548119. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The RCFD decoder is used in Range Computer CP-888-VPS-2, part of Radar Set AN/VPS-2. The RCFD decoder forms part of a binary to decimal decoder for 1.25 meter range increments, provides a single output called the E/L course output from the RCFD high and low data, and has circuits for detecting minimum and maximum range rates and velocity of less than 10 meters per second.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging data sheet number (see 5.1).

c. Selection of applicable level of preservation packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Requirement for first article submission.

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity).*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

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